RMO(INDIA) – 2004

Max Mark – 100 Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.

1. Consider in the plane a circle Γ with centre O and a line l not intersecting circle Γ. Prove that there is a point Q on the perpendicular drawn from O to the line l, such that for any point P on the line l, PQ represents the length of the tangent from P to the circle Γ.
2. Positive integers are written on all faces of a cube, one on each. At each corner (vertex) of the cube, the product of the numbers on the faces that meet at the corner is written. The sum of the numbers written at all the corners is 2004. If T denotes the sum of the numbers of all the faces, find all possible values of T.
3. Let α, β are the roots of the quadratic equation $x^2 + mx - 1 = 0$, where m is an odd integer. Let $\lambda_n = \alpha^n + \beta^n$ for $n \geq 0$, prove that for $n \geq 0$
   (a) λ_n is an integer.
   (b) $\gcd(\lambda_n, \lambda_{n+1}) = 1$.
4. Prove that the number of triples (A, B, C) where A, B, C are subsets of {1, 2, 3, …., n} such that $A \cap B \cap C = \phi$, $A \cap B \neq \phi$, $B \cap C \neq \phi$ is $7^n - 2 \times 6^n + 5^n$.
5. Let ABCD is a quadrilateral; X and Y be the mid points of AC and BD respectively and the lines through X and Y respectively parallel to BD and AC meet in O. Let P, Q, R, S be the midpoints of AB, BC, CD and DA respectively. Prove that:
   (a) Quadrilateral APOS and APXS have the same area.
   (b) The areas of quadrilateral APOS, BQOP, CROQ, and DSOR are all equal.
6. Let $< p_1, p_2, ..., p_n, ... >$ be a sequence of primes defined by $p_1 = 2$ and for $n \geq 2$, p_{n+1} is the largest prime factor of $p_1 p_2 ....p_n + 1$. (Thus $p_2 = 3, p_3 = 7$). Prove that $p_n \neq 5$ for every n.
7. Let x and y be positive real numbers such that $y^3 + y \leq x - x^3$. Prove that
   (a) $y < x < 1$.
   (b) $x^2 + y^2 < 1$.

******
****
**